

and use hermetically sealed compressors that are designed and integrated into the application cabinet to withstand vibration, dust and harsh climatic conditions.

I have come up with a cost-effective solution based on a pre-existing automotive solution. It comprises an existing automotive air-conditioner compressor and an AC induction motor for driving it. The motor can be controlled to operate for a select temperature required in the cabin. These compressors can sustain dust, high temperatures and vibration. Leakage of R-134A gas from the compressor is also almost impossible for applications in shelters. This serves the purpose in shelters as well, as it is a cost-effective solution.

What were the biggest design challenges faced while re-engineering the system for a military application?

In military applications, the compressor must withstand the following conditions:

High temperatures. According to MIL-STD 461E, the compressor should withstand up to 55°C. This temperature range is already available in a vehicle's engine with temperature running up to 100 degrees centigrade.

Dust test. The compressor is already proven to dust of all terrain conditions where the vehicles run around and still perform without any hindrance. This explains why I use it in my solution.

Leakage of R-134A gas. Freon gas is the standard used for refrigeration. Its grade is numbered and legally authorised to use in refrigeration. Earlier, it was R-12A for applications. Now it is mandatory to use R-134A. It is prone to leakage from capillaries, so care is taken in automotive systems to sustain all the vibrations and harsh climatic conditions. This has performed well all these years in our vehicles, which proves it is rugged and can sustain the required conditions in shelter applications.

Variation of RPM. In vehicles, RPM of the engine varies depending on the run of the vehicle and has a governing arrangement to sustain the stress. This is a boon for my application where I am running the compressor on an electric motor. Moreover, I am running the compressor only in standstill conditions. It is carried in a sheltered vehicle in turned off condition.

What kinds of design challenges did you face while addressing the power needs of shelters with this solution?

One condition for design and integration of AMF panels for DG sets used at high altitudes is that the DG set should be able to operate, sustain and survive at low temperatures of high altitudes. The main problem here is the availability of oxygen and very cold and dusty climate.

Specific to the usage of the DG set, I design monitoring circuitry dedicated to look after the operating temperature conditions of the engine. Starting the engine when the coolant and lubricating oil are in 'freeze' condition will damage the rings, piston and cylinder block. We have also added a monitoring circuitry that melts the liquids before the engine is started.

You must be incorporating microcontrollers to make such critical decisions?

Microcontroller is a must in such applications. Specifically talking about this critical decision, it is a simple thing that when starting the engine, first a microcontroller senses the temperature of the engine at its lubricating oil, coolant and fuel tank. The microcontroller cranks the engine only if the liquids are at a temperature where they will be in liquid state. Otherwise, it first turns on the heater plugs at all or required points. Once they have reached the set temperatures, the engine is cranked.

What MCUs do you recommend for these applications?

Usually, I select a single microcontroller with minimum configuration or architecture required for the application. Such applications can be catered with an AT89C52. Now I am trying to apply it with Raspberry Pi where I can play big logics, conditions and much smarter solutions.

How is the distribution of power managed? Also, do we see Raspberry Pi anywhere in the power electronics picture?

Use of Raspberry Pi in our projects is in our future scope and still in proposal stage. But we have started working on the applications as I have mentioned earlier. For instance, it could be a dedicated ruggedised computer module mountable in a 48cm (19-inch) rack. It goes with Linux and is very convenient to use in connectivity through Ethernet and solidstate data storage in real time with some external hardware. Also, it is a low-cost solution for my applications.

In power distribution and switch gear handling, Raspberry Pi has a big future in our organisation. We are mostly working on Python programming for our applications. Python is much simpler and allows fast development. Our engineers have done some creative work on it and are confident about it. Currently, we are focusing on the applications of power distribution from a PDB to be used in shelter.

Once the concept to product development stage is accomplished, what approach do you recommend for selection of components and modules in the design?

Every design engineer has his own hand and way of getting a solution for the requirement. Likewise, I go with my own approach. Component and module consideration starts at the concept stage itself. Depending on their availability, time to procure and the cost, which is the most important thing, I select components at the concept stage itself.

While considering modules, I look into two things. One is the performance in various applications and climatic conditions. Second is the module's time of existence. Sometimes a module could be a solution in some other application while also being suitable for my concept. Sometimes a module cannot be related to electronics but it still could be suitable in electronic applications. Consider, for example, the air-conditioner I was talking about earlier. ●

(The complete interview can be read at electronicsforu.com)